AERO 4442A: Transatmospheric and Space Propulsion: Gibbs Assignment



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INTRODUCTION

In the design of rocket engines, a thorough understanding of the post-combustion temperature and composition is essential. The performance of a rocket engine hinges on its combustion chamber temperature, T_c or for us T_{eq}, which directly influences performance characteristics. Furthermore, knowing the composition of the combustion products helps optimize the engine by indicating whether the combustion process has reached equilibrium, achieved complete combustion, and if there is still unused energy. For this reason, it is critical to accurately determine the post-combustion conditions for various fuel to oxidizer ratios, combustion chamber pressures p_c, and different types of fuels and oxidizers that may be used in rocket combustion.

This report investigates the relationship between the initial conditions—namely the fuel-oxidizer ratio (f/O) and combustion pressure p_c—and the resulting equilibrium state characterized by the post-combustion temperature T_{eq} and the composition of the exhaust products. The findings presented here could be used to guide design improvements and enhance performance characteristics for rocket engines.

NOMENCLATURE

a ..... Polynomial constants for species properties
α_i ..... Lagrange multipliers
$\bar{B}_i$ ..... Total number of atoms of type i in the mixture
β ..... Under-relaxation parameter
$(\tilde{cp})_k$ ..... Specific heat at constant pressure for species k
$\eta_{i,k}$ ..... Number of atoms of type i in species k
$\tilde{g}_k$ ..... specific Gibbs free energy for species k
H ..... Total mixture enthalpy
$\tilde{h}_k$ ..... Specific enthalpy for species k
N_A ..... Avogadro's number
n_a ..... Number of distinct atomic species
n_s ..... Number of species within mixture
p ..... Pressure
p_{ref} ..... Reference pressure
$\Re$ ..... Universal gas constant [8314 J/(kmol K)]
σ_k ..... Moles of species k
σ_m ..... Total moles within mixture
$\tilde{s}_k$ ..... Specific entropy for species k
T ..... Temperature (K)

RESULTS

When considering the combustion of hydrogen and oxygen, if complete combustion is assumed, the reaction would proceed as:

$$H_2 + \frac{1}{2}O_2 \rightarrow H_2O \tag{1}$$

The Gibbs free energy minimization technique described in the methods was used, here we solved across a range of equivalence ratios. Below in Figure 1, the equilibrium temperature reached proceeding combustion is plotted against these equivalence ratios. Note that when solving, we are assuming a combustion chamber pressure of $p_c = 10atm$, and a pre-combustion temperature of H_2 and O_2 of 298 [k] [1].

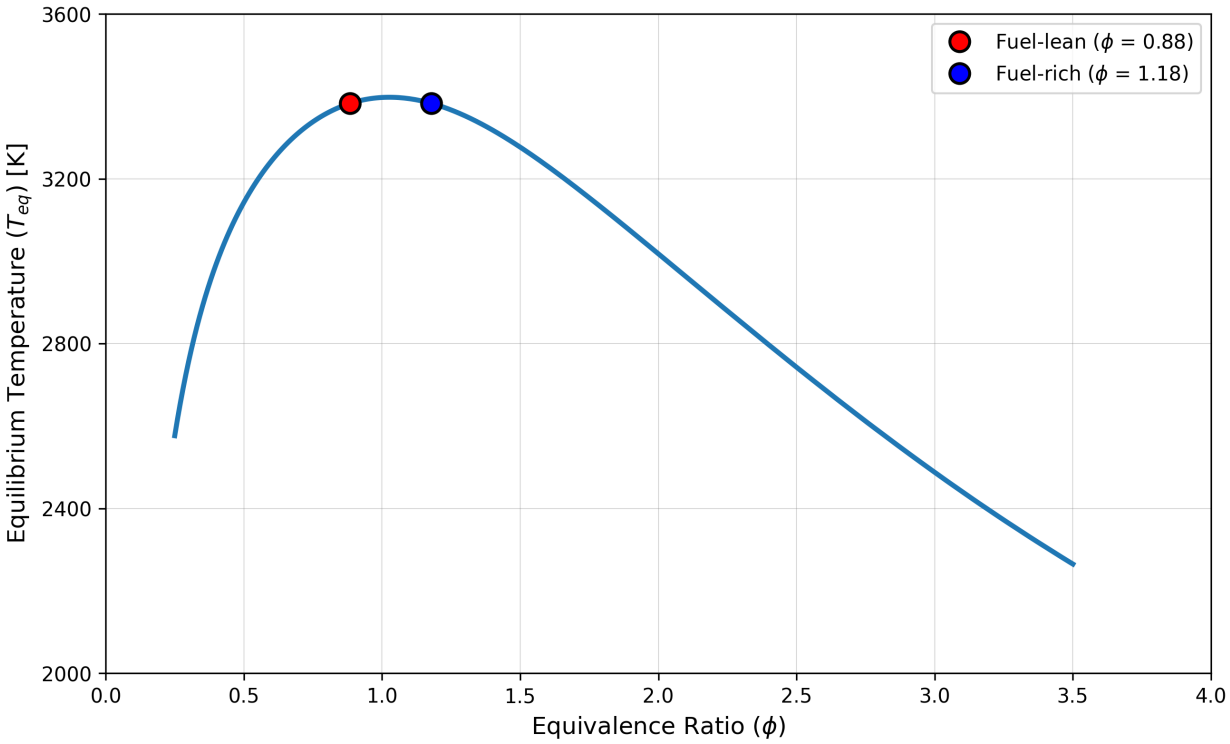


Figure 1: Temperature vs equivalence ratio ϕ

Highlighted in Figure 1 is where $T_{eq} = 3383[K]$. This occurs twice at $\phi = 0.88$, and $\phi = 1.18$. As can be seen above, T_{eq} reaches its peak, at $\phi = 1.03$, with an equilibrium temperature of $T_{eq} = 3397K$, this is good news for the results at a equivalence ratio of 0.88, and 1.18, as we are approaching the maximum equilibrium temperature, and in the context of propulsion the conditions that allow for the maximum thrust. For the two equivalence ratios that result in $T_{eq} = 3383[K]$ the reactions would proceed as follows (with the molar amounts reported to two decimal places).

For the fuel lean reaction ($\phi = 0.88$):

$$(1.35)H_2 + (0.77)O_2 \rightarrow (1.04)H_2O + (0.12)O_2 + (0.20)OH + (0.05)O + (0.17)H_2 + (0.07)H \tag{2}$$

And for the fuel rich reaction ($\phi = 1.18$):

$$(1.52)H_2 + (0.65)O_2 \rightarrow (1.06)H_2O + (0.03)O_2 + (0.14)OH + (0.02)O + (0.34)H_2 + (0.10)H \tag{3}$$

The above reactions show the products present with a mass fraction greater than 0.01%. The post-combustion mixture is shown below in Figure 2. The molar masses for all of the species can be seen in the appendix of this report.

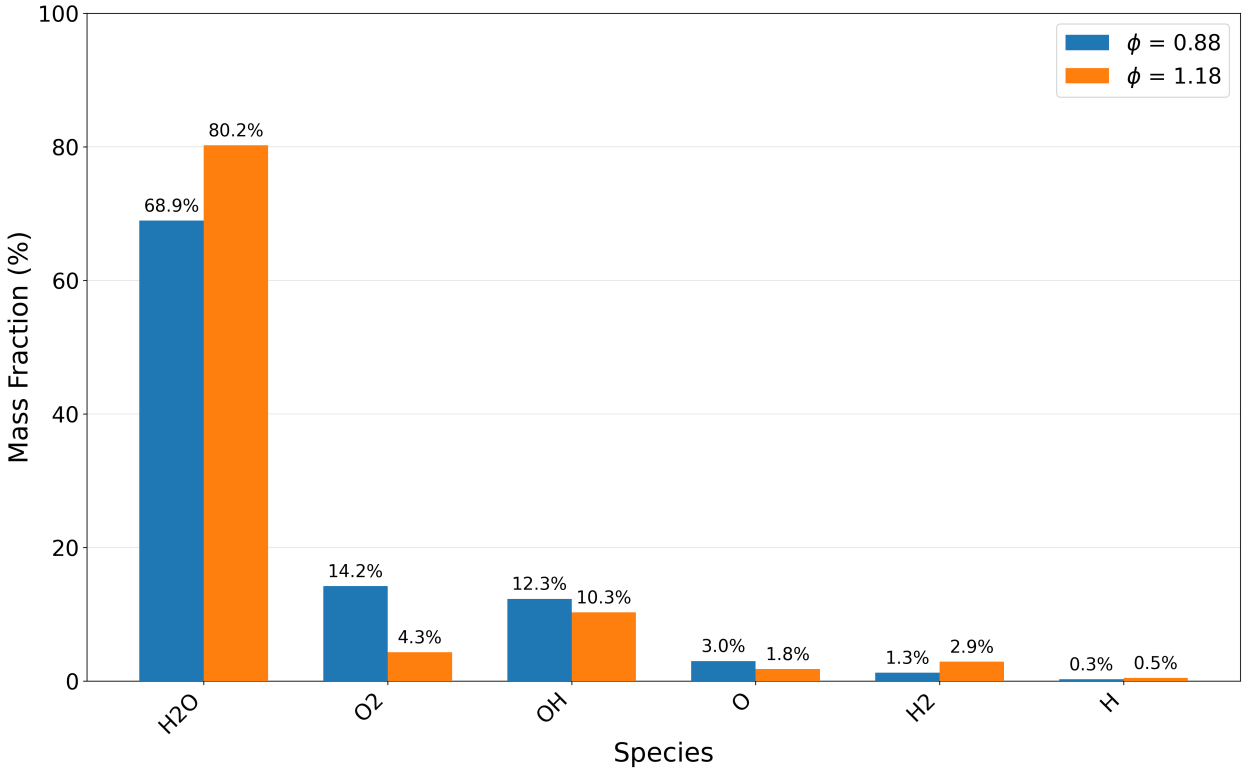


Figure 2: Species mass fractions

H_2O predominantly dictates the product composition under both scenarios, being more prevalent in the fuel-rich equivalence ratio, at 80.2% compared to 68.9% when $\phi = 0.88$. The fuel-lean mixture retains a significant proportion of unreacted O_2, amounting to 14.2%. In both scenarios, the manifestation of OH radicals is seen, attributable to high-temperature dissociation processes. These variations underscore the influence of the equivalence ratio in changing the equilibrium composition while sustaining a consistent temperature T_{eq} (3383K), with fuel-rich conditions showing greater water production and fuel-lean conditions fostering an excess of oxygen alongside augmented dissociation byproducts [2].

We can also examine the effects increased combustion chamber pressure has on T_{eq}. Figure 3 below shows the same results as Figure 1, but plotted over various values of p_c.

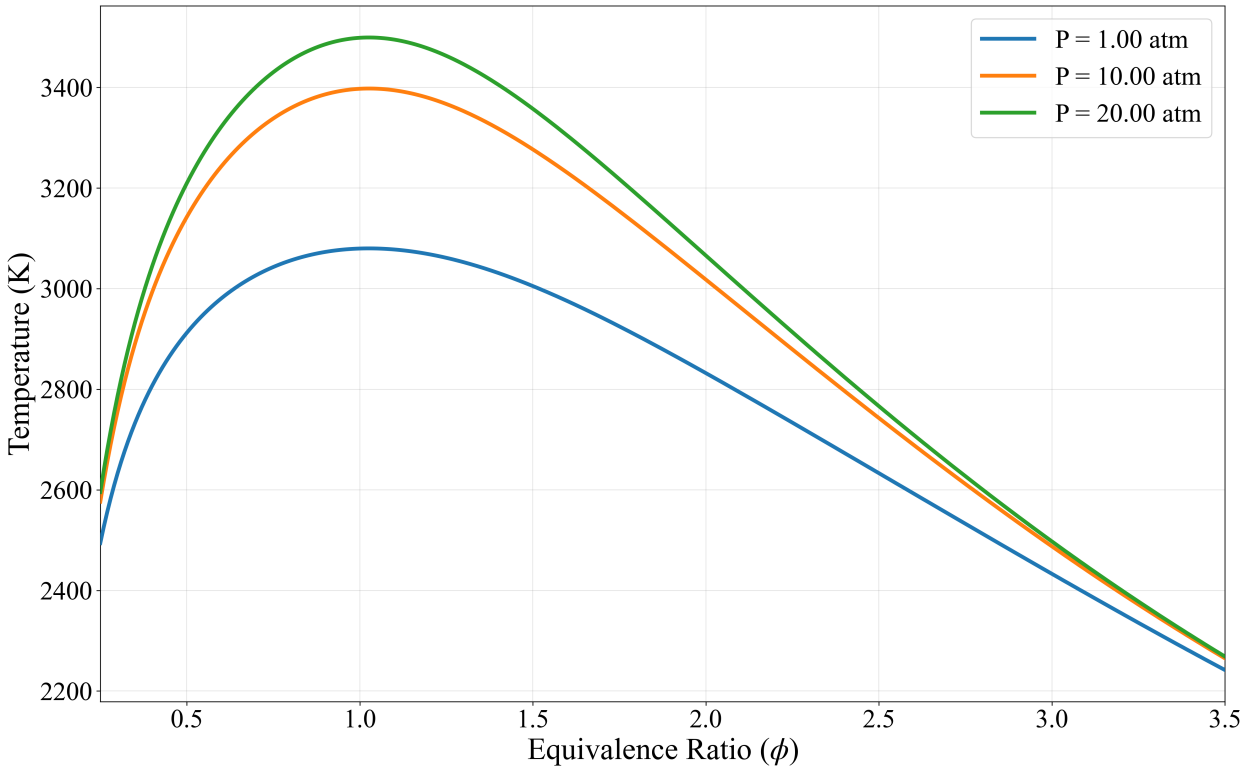


Figure 3: Temperature vs equivalence ratio ϕ at different pressures

The increased combustion chamber pressure does lead to higher T_{eq} values, especially at equivalence ratios near unity. This effect diminishes as we reach fuel lean or rich mixtures.

METHODOLOGY

This study employs a numerical simulation approach using the "Gibbs free energy minimization technique" to evaluate the post-combustion conditions present at equilibrium given certain pre-combustion conditions (type of fuel and oxidizer, equivalence ratio, ϕ, combustion pressure p_c, and temperature). The methods follow those outlined in the course text [3]. The results focus on the post-combustion temperature T_{eq} and composition of the products present post-combustion. The simulation is implemented in Python, the implementation consists of two main parts, (1) model prepossessing to determine the initial conditions for various equivalence ratios (ϕ), and (2) a Gibbs energy minimization solver to determine the equilibrium conditions post-combustion. The theory, and methodology followed in the model is outlined below.

Stoichiometry and Equivalence Ratio

For a specific reaction, we can consider the fuel to oxidizer ratio (f/o), this represents the ratio of moles of the fuel to the moles present in the oxidizer. When considering the stoichiometric complete combustion of hydrogen and oxygen, the reaction can be written as:

$$H_2 + \frac{1}{2}O_2 \rightarrow H_2O \tag{4}$$

In the stoichiometric case, where H_2 is the fuel and O_2 is the oxidizer, this provides a fuel to oxidizer ratio on a molar basis of $(f/o)_{st} = \frac{1}{0.5} = 2$. However, it is also possible that the reaction could proceed as:

$$H_2 + O_2 \rightarrow H_2O + O \tag{5}$$

Yielding a (f/o) ratio of 1.

At this point we may introduce the equivalence ratio (ϕ), this is the ratio, of the actual (f/o) ratio that occurred, with the stoichiometric fuel to oxidizer ratio $(f/o)_{ST}$ when considering the two reactions above.

$$\phi = \frac{(f/o)}{(f/o)_{st}} = \frac{\frac{1}{1}}{\frac{1}{0.5}} = \frac{1}{2} \tag{6}$$

In reality, we do not know the products of the reaction, however, in practice we can control the initial conditions so that we have desired post-combustion conditions. In order to do this we must consider all possible products that could occur from a given reaction. In fact, from the combustion of hydrogen and oxygen, we consider all of the species in Table 1, to be a possible product.

Table 1: Possible products

Species	**Molecular Weight**
H_2	2.0159
O_2	31.9988
H	1.0079
O	15.9994
OH	17.0073
H_2O	18.0153
H_2O_2	34.0147
HO_2	33.0067

Given a set of initial conditions (equivalence ratio), we can proceed to determine the products of the reaction (σ_k), and the temperature of the reaction at equilibrium (T_{eq}) using the Gibbs minimization technique.

Gibbs free energy

Defining the Gibbs energy as G, also referred to as the chemical potential energy, it is given as the difference between the enthalpy (H), and the products of the absolute temperature (T), and entropy (S).

$$G = H - TS, \quad [J] \tag{7}$$

When considering a mixture, with k species, the Gibbs energy can be found as 8, where $\tilde{g_k}$ is the Gibbs energy per mole of species k, and σ_k is the moles of species k present.

$$G = \sum_k \tilde{g}_k \sigma_k \tag{8}$$

We note that at equilibrium, when the change in pressure and temperature has stopped, the Gibbs free energy will be at its minimum, and we can write.

$$\sum_k \tilde{g_k} d\sigma_k = 0 \tag{9}$$

If we can solve for this equilibrium condition in 9, when the change in $d\sigma_k = 0$, we can use a series of equations, to be outlined below, to solve iteratively until the equilibrium conditions are met. This solution will represent the post-combustion equilibrium conditions.

Iterative solution

Understanding that the solution we are looking for occurs when the Gibbs free energy is minimized, a series of equations can be used to find this solution. Since we are trying to find the minima of a function with respect to set constraints, the method of Lagrange multipliers can be used. The resulting series of equations will have $n_s + n_a + 2$ total number of equations, recalling that $n_s and n_a$ represent the number of species and atoms present in the mixture.

When considering the combustion of hydrogen and oxygen, while wanting to consider all possible products that form from Table 1, we have $n_s = 8$ and $n_a = 2$. To deal with the size of the system, the Newton-Raphson technique can be utilized. This ultimately results in a relatively simple process of finding equilibrium, as outlined below. The derivation of these equations can be seen in [3].

The results of utilizing the Lagrange multipliers, and Newton-Raphson method allow us to write the series of equations in matrix form 10.

$$[A][X] = [C] \tag{10}$$

Where:

$$A = \begin{bmatrix} \sum_k \eta_{1,k}\sigma_k\eta_{1,k} & \sum_k \eta_{1,k}\sigma_k\eta_{2,k} & \cdots & \sum_k \eta_{1,k}\sigma_k\eta_{\eta,k} & \sum_k \eta_{1,k}\sigma_k & \sum_k \eta_{1,k}\sigma_k \frac{\tilde{h_k}}{\Re T} \\ \sum_k \eta_{2,k}\sigma_k\eta_{1,k} & \sum_k \eta_{2,k}\sigma_k\eta_{2,k} & \cdots & \sum_k \eta_{2,k}\sigma_k\eta_{\eta,k} & \sum_k \eta_{2,k}\sigma_k & \sum_k \eta_{2,k}\sigma_k \frac{h_k}{\Re T} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \sum_k \eta_{\eta,k}\sigma_k\eta_{1,k} & \sum_k \eta_{\eta,k}\sigma_k\eta_{2,k} & \cdots & \sum_k \eta_{\eta,k}\sigma_k\eta_{\eta,k} & \sum_k \eta_{\eta,k}\sigma_k & \sum_k \eta_{\eta,k}\sigma_k \frac{\tilde{h_k}}{\Re T} \\ \sum_k \eta_{1,k}\sigma_k \frac{\tilde{h_k}}{\Re T} & \sum_k \eta_{2,k}\sigma_k \frac{\tilde{h_k}}{\Re T} & \cdots & \sum_k \eta_{\eta,k}\sigma_k \frac{\tilde{h_k}}{\Re T} & \sum_k \sigma_k \frac{\tilde{h_k}}{\Re T} & \sum_k \sigma_k \left[\frac{(\tilde{c_p})_k}{\Re} + \frac{\tilde{h_k}}{\Re T}^2\right] \\ \sum_k \eta_{1,k}\sigma_k & \sum_k \eta_{2,k}\sigma_k & \cdots & \sum_k \eta_{\eta,k}\sigma_k & \sum_k \sigma_k - \sigma_m & \sum_k \sigma_k \frac{\tilde{h_k}}{\Re T} \end{bmatrix} \tag{11}$$

$$C = \begin{bmatrix} \bar{B_1} + \sum_k \eta_{1k}\sigma_k \left(\frac{\tilde{g_k}}{\Re T} - 1\right) \\ \bar{B_2} + \sum_k \eta_{2k}\sigma_k \left(\frac{\tilde{g_k}}{\Re T} - 1\right) \\ \vdots \\ \bar{B_{na}} + \sum_k \eta_{na,k}\sigma_k \left(\frac{\tilde{g_k}}{\Re T} - 1\right) \\ \frac{H}{\Re T} + \sum_k \sigma_k \frac{\tilde{h_k}}{\Re T} \left(\frac{\tilde{g_k}}{\Re T} - 1\right) \\ \sigma_m + \sum_k \sigma_k \left(\frac{\tilde{g_k}}{\Re T} - 1\right) \end{bmatrix} \quad X = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_{na} \\ \Delta \ln \sigma_m \\ \Delta \ln T \end{bmatrix} \tag{12}$$

These are the core of the solution, we will populate these matrices based on our initial conditions (equivalence ratio), solve for $[X]$, and find the new values of σ_k, σ_m and T. We will continue to solve iteratively until the solution converges at which point equilibrium has been met.

Initial conditions

Firstly, we must describe some of the unknowns that are needed to populate the matrix. For us, $\bar{B_i}$, and H are constants, since we know the quantities of the reactants at start, the equivalence ratio and temperature pre-combustion are assumed to be $298[K]$. $\bar{B_i}$ is the total number of atoms i contained in the mixture at all times. Since this is conserved throughout the reaction, when calculating this we use the fuel to oxidizer ratio that has been specified by the equivalence ratio. Here η_{ik} is the number of atoms of atom i in species k.

$$\bar{B_i} = \frac{B_i}{N_A} = \sum_k \eta_{i,k}\sigma_k \tag{13}$$

$$B_i = \sum_k N_A \eta_{i,k}, \sigma_k \tag{14}$$

H represents the thermal energy contained in the system. In this case it also represents the total energy in the system since we are considering a system with no change in height, closed volume, with negligible velocity, and there is no work added/subtracted from the flow. We should also note here that the polynomials $a_0, a_1, a_2...$ are constants for functions of thermodynamic properties calculated at a specific reference temperature and pressure; $(T_{ref} = 293, p_{ref} = 1bar)$. They are taken from [3]. The table can be found in the appendix of this document. $\Re$ is the universal gas constant.

$$H = \sum_k \sigma_k \tilde{h}_k \tag{15}$$

Where the molar enthalpy can be written as:

$$\tilde{h_k} = \Re T \left(a_0 + a_1 \frac{T}{2} + a_2 \frac{T^2}{3} + a_3 \frac{T^3}{4} + a_4 \frac{T^4}{5} + a_5 \frac{1}{T} \right) \tag{16}$$

The rest of the values used to populate the matrices in (10), are found from an initial guess. In the simulation the initial conditions follow (17), as outlined in [3]. For the hydrogen and oxygen reaction considered in this report, this yields the initial values for σ_k for all species to be 0.0125.

$$\sigma_m[kmol] = 0.1 \quad \sigma_k[kmol] = 0.1/n_s \quad T_{eq}[K] = 7800 \tag{17}$$

The last values needed to populate and solve (10) are $\tilde{h_k}, \left(\tilde{c_p}\right)_k$ and $\tilde{g_k}$. The molar enthalpy can again be found using (16). The remainder can be found from the following equations.

$$\left(\tilde{c_p}\right)_k = \Re \left(a_0 + a_1 T + a_2 T^2 + a_3 T^3 + a_4 T^4 \right) \tag{18}$$

$$\tilde{g}_k(T,p) = \tilde{h}_k(T,p_{\text{ref}}) - T\tilde{s}_k(T,p_{\text{ref}}) + RT \ln\left(\frac{\sigma_k}{\sigma_m}\right) + RT \ln\left(\frac{p}{p_{\text{ref}}}\right) \tag{19}$$

Where for (19), we need the molar entropy $\tilde{s_k}$:

$$\tilde{s_k} = \Re \left(a_0 lnT + a_1 T + a_2 \frac{T^2}{2} + a_3 \frac{T^3}{3} + a_4 \frac{T^4}{4} + a_6 \right) \tag{20}$$

Under relaxation parameter

Finally we have populated the matrix in (10) after solving for $[X]$, the following procedure must be followed to maintain stability [4, 5]. For species where the kmol amount exceeds 1×10^{-8} and $\Delta \ln(\sigma_k)$ is positive, the relaxation parameter β_1 is computed as:

$$\beta_1 = \frac{2}{\max\left(\left|\Delta \ln(T)\right|, \left|\Delta \ln(\sigma_m)\right|, \left|\Delta \ln(\sigma_k)\right|\right)} \tag{21}$$

For the remaining species with positive $\Delta \ln(\sigma_k)$ values, β_2 is computed as:

$$\beta_2 = \frac{\ln(1 \times 10^{-4}) - \ln(\sigma_k) + \ln(\sigma_m)}{\Delta \ln(\sigma_k) - \Delta \ln(\sigma_m)} \tag{22}$$

The final under-relaxation parameter β is then determined as:

$$\beta = \min\left(1, \beta_1, \beta_2\right) \tag{23}$$

The relaxation parameters are then used to update the values of (T, σ_m, σ_k) as follows:

$$\ln(T_{\text{new}}) = \ln(T_{\text{old}}) + \beta \Delta \ln(T) \tag{24}$$

$$\ln(\sigma_{m,\text{new}}) = \ln(\sigma_{m,\text{old}}) + \beta \Delta \ln(\sigma_m) \tag{25}$$

$$\ln(\sigma_{k,\text{new}}) = \ln(\sigma_{k,\text{old}}) + \beta \Delta \ln(\sigma_k) \tag{26}$$

Where to obtain the solution values for σ_k, we use the results of the matrix $[X]$ and the following equation:

$$\Delta \ln \sigma_k = \Delta \ln \sigma_m + \left(\frac{\tilde{h_k}}{\Re T}\right) \Delta \ln T - \frac{\tilde{g_k}}{\Re T} + \sum_i \alpha_i \eta_{i,k} \tag{27}$$

After each iteration, the solution values of σ_k, σ_m, and T are used to calculate the new values of $\sigma_k, \sigma_m . T, \tilde{g_k}, (\tilde{c_p})_k$, and $\tilde{h_k}$. These are in turn used to repopulate the matrix, this process continues until the solutions converge. The solution represents the resulting equilibrium conditions T_{eq} and σ_k, where the Gibbs free energy has been minimized.

APPENDIX

Table 2: Polynomial Coefficients for Different Species

Species	a_0	a_1	a_2	a_3	a_4	a_5	a_6
H_2	2.932866	8.266080e-04	-1.460000e-07	1.540000e-11	-6.890000e-16	-813.065597	-1.024329
O_2	3.660961	6.563660e-04	-1.410000e-07	2.060000e-11	-1.300000e-15	-1215.977250	3.415362
H_2O	2.677038	2.973183e-03	-7.740000e-07	9.440000e-11	-4.270000e-15	-29885.893800	6.882556
OH	2.838646	1.107256e-03	-2.940000e-07	4.210000e-11	-2.420000e-15	3943.958520	5.844527
H	2.500003	-5.650000e-09	3.630000e-12	-9.200000e-16	7.950000e-20	25473.658900	-0.446698
O	2.543637	-2.730000e-05	-4.190000e-09	4.950000e-12	-4.800000e-16	29226.012000	4.922295
H_2O_2	4.573335	4.049841e-03	-1.290000e-06	1.970000e-10	-1.130000e-14	-18054.812100	0.704278
HO_2	4.172287	1.881176e-03	-3.460000e-07	1.950000e-11	1.760000e-16	61.810296	2.957677
H_2CO	3.169527	6.190000e-03	-2.250000e-06	3.660000e-10	-2.200000e-14	-14500.000000	6.040000
CH_4	1.635526	1.010000e-02	-3.370000e-06	5.350000e-10	-1.715519e+01	-10000.000000	-9.990000
CO	3.048486	1.350000e-03	-4.860000e-07	7.890000e-11	-4.700000e-15	-14300.000000	6.070000
CO_2	4.636595	2.740000e-03	-9.960000e-07	1.600000e-10	-9.160000e-15	-49000.000000	-1.940000
N_2	2.950000	1.400000e-03	-4.930000e-07	7.860000e-11	-4.610000e-15	-92400.000000	5.870000
NO	3.260000	1.190000e-03	-4.290000e-07	6.940000e-11	-4.030000e-15	9920.000000	6.370000
N_2O	4.820000	2.630000e-03	-9.580000e-07	1.600000e-10	-9.770000e-15	8070.000000	-2.200000

Table 3: Molecular Weights of Different Species

Species	**Molecular Weight (g/mol)**
H_2	2.0159
O_2	31.9988
H	1.0079
O	15.9994
OH	17.0073
H_2O	18.0153
CO	28.0101
CO_2	44.0095
N_2	28.0134
NO	30.0061
N_2O	44.0128

Data and code availability. Model input data, code, and results can be found at: https://github.com/DavidMacKay709/Gi

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